

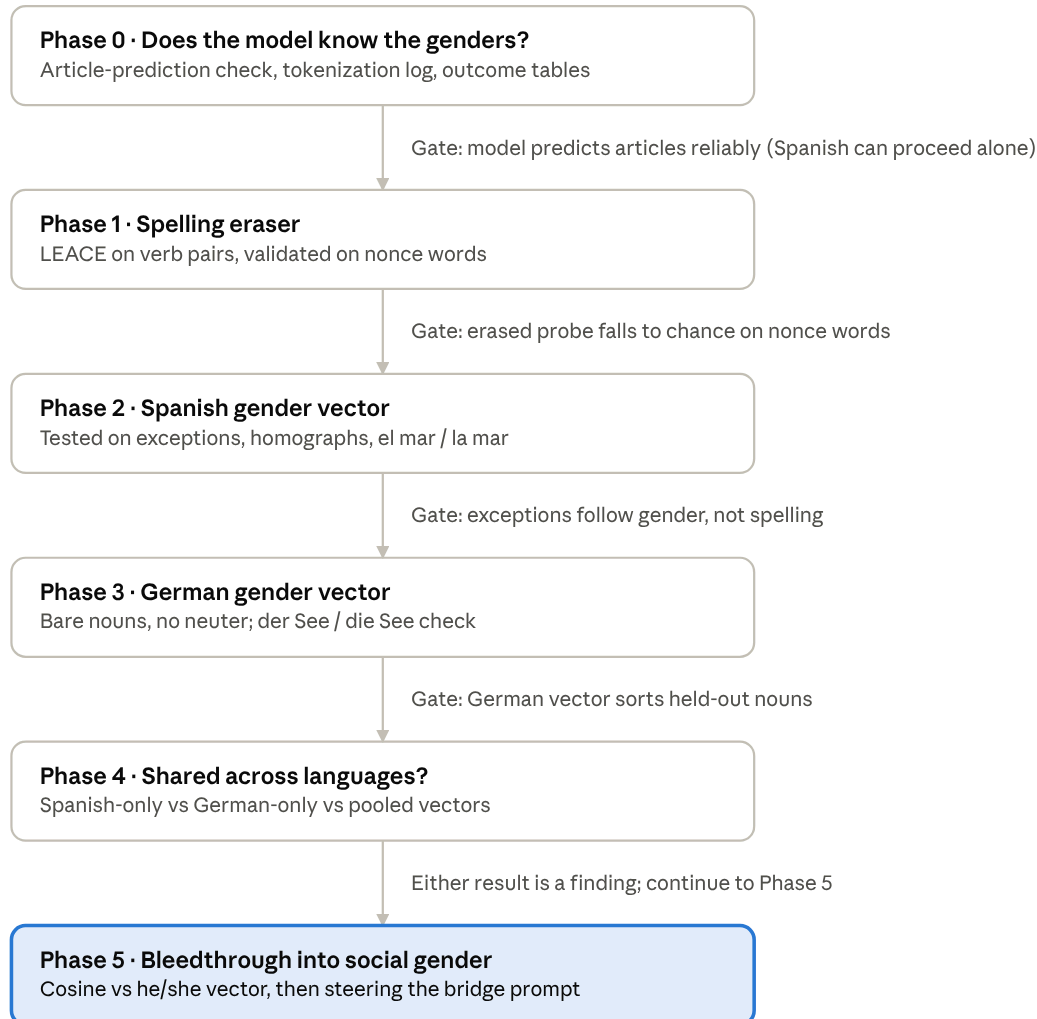
Grammatical Gender Bleedthrough in LLMs: Project Design

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Does a model's representation of grammatical gender, learned from inanimate nouns, overlap with its representation of social gender? Five gated phases, each producing a usable result even if the next never runs.

At a glance

Each phase gates the next, and each stops with a result



project roadmap · 6 phases, 5 gates

If a gate fails, the tables in Phases 1 and 2 say where to go back. Phases 0-2 produce a usable result even if nothing after them runs.

Question and null hypothesis

Question: Does grammatical gender, which is arbitrary for inanimate nouns, shape the model's language-agnostic representation of those nouns, and does it overlap with social gender? This is the Boroditsky bridge effect, tested inside a model instead of in people.

Null (comparative, not absolute): Which language assigns a noun which gender predicts nothing about the model's shared representation of that noun. No claim is needed about whether the "true" bridge is genderless.

Test set for the null: nouns whose gender flips between German and Spanish, for example *die Brücke / el puente, der Mond / la luna, die Sonne / el sol, der Schlüssel / la llave*.

Scope decisions: Spanish and German only; German neuter excluded; binary masculine vs feminine throughout.

Setup

Prototype on the Mac; rent a GPU only for larger-model extraction runs; do all analysis locally.

	Mac (M-series, 16 GB unified)	RunPod (rented GPU)
Use for	Stimulus building, pipeline code, Phases 0-1 end to end, small-model results	Extraction on larger models, with a script already debugged at home
Model size	2-3B in half precision (plan for ~10-12 GB usable)	Whatever the rented card fits
Backend	PyTorch on MPS (not MLX; the interp tooling is PyTorch)	PyTorch on CUDA
Session pattern	Open-ended	Extract, save activations to disk, download, shut down

Software: PyTorch, Hugging Face transformers, TransformerLens or nnsight for hooks, EleutherAI's concept-erasure package for LEACE, scikit-learn for probes.

Model choice: a small open multilingual model. Pick the one that passes the Phase 0 behavioral check best, not the one with the best benchmark scores.

Rules:

- No heavy quantization (4-bit); it distorts the activations you're measuring.
- Never compare activations extracted on different hardware. For any one model, extract everything on one device.
- Record activations at the last token of each target word, across all layers.

Phase 0: Does the model know the genders?

There's no point looking for a representation of something the model gets wrong. Small models can be shaky in German.

1. **Behavioral check.** For every noun in your lists, prompt so the next token is the article and compare the probabilities of *el/la* and *der/die*. Record per-noun accuracy.
2. **Drop or flag** nouns the model gets wrong; they can't anchor a gender vector.
3. **Check tokenization** of every stimulus. Note which nouns and nonce words split into several tokens.
4. **Write the outcome tables** for each phase before running it: possible results, what each means, what you'd do next. If two outcomes would lead to the same conclusion, the experiment isn't discriminating.

Gate: if accuracy is poor for German, try a different model before continuing; Spanish-only Phases 1–2 can proceed regardless.

Phase 1: Build and validate a spelling eraser

In Spanish, *-a/-o* correlates strongly with gender, so a gender vector risks being a spelling detector. Train an eraser on words where the ending carries no gender.

Train: LEACE on regular *-ar* verb pairs (*hablo/habla, canto/canta*). Same stem, ending marks person, not gender. Never train on nouns or adjectives: their *-a/-o* is gender.

Validate on nonce words (*brelda/brelpo*), presented bare:

1. Record activations at the last token.
2. Train a logistic-regression probe for *-a* vs *-o*. Split train/test **by stem**.
3. Apply the eraser, then train a **fresh** probe on the erased activations.

Result after erasure	Meaning	Next
Probe accuracy near chance	Eraser removes -a/-o in general	Proceed to Phase 2
Probe accuracy stays high	Eraser learned person, not form	Add half the nonce stems to training; retest on the other half
Still high after that	Spanish-specific confounds	Add Italian (<i>parlo/parla</i>) and Portuguese (<i>falo/fala</i>) verbs

Caveat: if *brel*da tokenizes as *brel* + *da*, the ending sits in a separate token, so high pre-erasure accuracy is trivial. The erasure comparison is still valid.

Phase 2: Spanish grammatical-gender vector

Extract a gender direction from regular nouns, then test it where spelling and gender disagree.

Train: regular inanimate nouns, bare (no article), using the **erased** activations from Phase 1. Difference of class means or a linear probe.

Test set 1: exceptions, scored by subgroup. Held out, never used for training.

- -ma Greek borrowings: *el problema*, *el tema*, *el sistema*, *el programa*
- Clippings: *la foto*, *la moto*
- True exceptions: *el día*, *la mano*

Test set 2: noun/verb homographs, disambiguated with gender-neutral context: *mi camino* vs *yo camino*, *mi cuenta* vs *él cuenta*. Same string; gender should appear only in the noun reading.

Test set 3: same word, both genders: *el mar* / *la mar*. This tests gender from context, so measure at the noun token.

Result on exceptions	Meaning	Next
Follow grammatical gender	Gender is encoded beyond surface form. Not yet social gender.	Phases 3–5
Follow spelling	Spelling detector	Revisit Phase 1

Result on exceptions	Meaning	Next
<i>-ma</i> and clippings right, <i>día/mano</i> wrong	Vector captures form rules and word relations, but misses gender the model clearly knows (it produces those articles correctly)	Try other layers; the information is somewhere else
Nothing sorts cleanly	Can't find it with this instrument; not evidence of absence	Layer sweep, nonlinear probe, more data

Phase 3: German gender vector

German gender is much less predictable from spelling, so no eraser is needed. The main leak is the article, which bare nouns avoid.

- **Train:** masculine and feminine inanimate nouns, bare. No neuter.
- **Watch for reliable suffixes** (*-ung*, *-heit*, *-keit*, *-schaft* feminine). Keep them as a separate test subgroup so a suffix rule can't pass for gender.
- **Spelling-constant check:** *der See* (lake) / *die See* (sea). Identical string, opposite gender.

Phase 4: Is grammatical gender shared across languages?

Three vectors, each tested on held-out nouns from both languages, including the flipped pairs.

Vector	Trained on
Spanish-only	Phase 2
German-only	Phase 3
Pooled	Spanish + German masculine vs Spanish + German feminine

Result	Meaning
Pooled vector wins on both languages	A language-independent notion of grammatical gender exists
Each single-language vector wins on its own language, fails on the other	Gender is encoded per language; a finding, not a failure
Pooled is mediocre at both	Directions differ enough that averaging blurs them

The flipped pairs test the null directly: in shared middle layers, does *bridge* lean feminine when read as *Brücke* and masculine when read as *puente*?

Phase 5: Does grammatical gender bleed into social gender?

The original question. Two vectors, one comparison, one intervention.

1. **Social-gender vector** from English people pairs: *he/she, man/woman, king/queen*.
2. **Cosine similarity** between it and your grammatical-gender vector, layer by layer. Near zero = grammatical gender stays grammatical. Aligned = something has bled.
3. **Steering**: inject the grammatical-gender vector (built only from inanimate nouns) into "The bridge was ____". Compare probabilities of preset adjective lists, *elegant, beautiful, slender vs sturdy, strong, solid*.
4. **Baseline**: a random vector of the same size, so you know a shift isn't just from perturbing the model.

Interpretation: the human effect failed to replicate (Mickan et al. 2014), so a null here is informative: either the effect isn't in the text, or the model doesn't pick it up. A positive result means a vector with no people in it moves people-stereotyped adjectives.

Stimulus lists to build

Most of the project's time goes here, not into compute.

- ☐ Regular *-ar* verb pairs, 1sg / 3sg (Phase 1 eraser)
- ☐ A few hundred Spanish-plausible nonce stems with both endings, checked against a dictionary (Phase 1 validation)
- ☐ Optional: Italian and Portuguese *-o/-a* verb pairs (Phase 1 fallback)

- ☐ Regular Spanish inanimate nouns, masculine -o and feminine -a (Phase 2 training)
- ☐ Spanish exceptions by subgroup: -ma, clippings, true exceptions (Phase 2 test)
- ☐ Noun/verb homographs with gender-neutral frames (Phase 2 test)
- ☐ German masculine and feminine inanimate nouns, suffix-marked ones tagged (Phase 3)
- ☐ German-Spanish flipped-gender pairs (Phases 4–5)
- ☐ English people pairs for the social-gender vector (Phase 5)
- ☐ Two adjective lists for the steering test (Phase 5)
- ☐ Tokenization log for every list above (Phase 0)

Pitfalls and open questions

Pitfalls

- **Erasing gender by accident.** Any eraser trained on data where -a/-o correlates with gender removes gender too.
- **Article leakage.** An article before a noun announces its gender; use bare nouns unless the test is about context.
- **Linear-only tools.** LEACE and linear probes can't see nonlinear encodings. "Not found" means not found linearly.
- **Stem leakage.** Always split train and test by stem, never by word.
- **Overclaiming Phase 2.** Encoding grammatical gender isn't the same as overlapping social gender; that's Phase 5.

Open questions

- Has Kann (2019) already answered part of this at the embedding level? Read before building Phases 2–5.
- Which small multilingual model passes Phase 0 best in German?
- Which layers count as "shared"? Decide from the Phase 4 results, not in advance.